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Hydrogen Transportation with Pipeline – Challenges and Path Forward

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Abstract

Hydrogen is increasingly recognized as a key energy carrier in global decarbonization strategies, offering clean combustion and flexible storage capabilities. As energy operators realize the potential of hydrogen as an energy carrier, repurposing existing natural gas pipeline infrastructure emerged as a promising alternative to constructing new hydrogen-dedicated networks. This paper investigates the suitability of current pipeline standards and guidelines for hydrogen service, with a focus on identifying technical gaps that must be addressed to enable safe and effective repurposing.

The methodology involves a comparative evaluation of widely adopted standards such as ASME B31.12, IGEM/TD/1, CGA G-5.6, and DVGW G 409, alongside hydrogen-specific initiatives like H2Pipe and PRCI collaboration with ASME. Key areas of assessment include hydrogen embrittlement, fatigue and fracture mechanics, qualification testing, and engineering critical assessments. Material compatibility is assessed through literature and technical references, examining the behavior of carbon and alloy steels under hydrogen exposure. Subsea pipeline considerations are also addressed, highlighting unique operational challenges and the conservative nature of existing design practices.

Case studies and pilot projects are incorporated to provide some context on how these standards are applied in practice and to identify limitations. The paper concludes with recommendations for future work, including the need for long-term field trials, improved inspection technologies, and refined material specifications tailored for hydrogen environments.

Results indicate that while pipelines are the most cost effective option for regional hydrogen transport, especially when repurposed, significant technical and regulatory gaps remain. These include material qualification, fracture control, and subsea applications, which must be addressed to ensure safe and scalable deployment.

Introduction

Hydrogen is increasingly recognized as a key energy carrier in global decarbonization efforts, with projections indicating it could meet 15% of global energy demand by 2050 [1]. Historically used in industries such as food processing, petrochemical refining, and ammonia production [2], hydrogen's role is shifting from feedstock to fuel. Its emergence as an energy carrier began during the 1970s oil crisis and gained momentum with the Kyoto Protocol in 1997 [3]. Hydrogen's clean combustion, producing

only water vapor, and its ability to be stored for future use make it attractive compared to intermittent renewables. As a result, energy operators are investing in infrastructure for hydrogen production, transport, and integration. However, hydrogen's small atomic size and low volumetric energy density pose engineering challenges, requiring specialized infrastructure for safe and efficient handling. Repurposing existing natural gas pipelines offers a cost-effective alternative to building new hydrogen-dedicated networks, especially in urban areas where construction is disruptive. These pipelines also benefit from long-standing public acceptance [4]. Using literature, this paper reviews hydrogen transportation, identifies gaps in existing standards and guidelines for pipeline repurposing, examines material compatibility under hydrogen exposure, addresses subsea pipeline considerations, and outlines future research directions.

Methodology

The approach includes literature review to identify optimal transport methods and an evaluation of standards related to repurposing natural gas pipelines for hydrogen transmission. Gaps, where identified, will then be bridged using available literature and initiatives.

Overview of Hydrogen Transport Methods

Hydrogen can be transported using conventional methods similar to oil and gas, including land transport, pipeline networks, and shipping. However, its unique physical and chemical properties must be considered when selecting the most suitable method. Factors such as transport quantity, distance, and local infrastructure influence this choice [5]. In their article, [5] examined various hydrogen storage and transport methods, focusing on technical and economic feasibility. This paper emphasizes transportation; storage is mentioned only when relevant.

According to [6], hydrogen can be moved as liquid hydrogen (LH_2), compressed hydrogen (CGH_2), ammonia (NH_3), or liquid organic hydrogen carriers (LOHC). Each option has benefits and limitations. LH_2 suits bulk shipping but requires cryogenic infrastructure and faces boil-off losses. CGH_2 is simpler but less efficient over long distances due to low energy density. NH_3 and LOHC offer safer and more stable conditions but need substantial energy to release hydrogen at the destination. The choice depends on distance, infrastructure, and intended use [6].

Standards and Guidelines Evaluation

A comparative evaluation of existing pipeline standards and hydrogen-specific guidelines is conducted to identify gaps and opportunities for repurposing. Some of the documents reviewed as part of this paper include:

- ASME B31.12 – Hydrogen Piping and Pipelines [7]
- ASME B31.8 – Gas Transmission and Distribution Piping Systems [8]
- ASME B31.4 - Pipeline Transportation Systems for Liquids and Slurries [9]
- Phase 2 of H2Pipe - Joint industry project (JIP) [10]
- IGEM/TD/1 - Steel pipelines for high pressure gas transmission [11]
- CGA G-5.6 - Hydrogen Pipeline Systems [12]
- DVGW G 409 - Conversion of high pressure gas steel pipelines for a design pressure of more than 16 bar for transportation of hydrogen [13]

This evaluation focuses on how these standards address hydrogen-specific challenges such as:

- Service-specific requirements

- Hydrogen embrittlement
- Fatigue and fracture mechanics
- Engineering Critical Assessments (ECA) – Refer to the case study by API 1104 Hydrogen Task Group
- Qualification testing and sampling procedures

The goal is to assess how well current standards support repurposing and where more guidance is needed. Furthermore, this study attempts to bridge gaps in current standards through insights from available literature and ongoing industry initiatives, such as the PRCI collaboration with ASME and the DNV-led Joint Industry Projects (JIPs).

Material Compatibility and Mechanical Behavior

The feasibility of transporting hydrogen through existing pipeline steels has been extensively studied. Research [14] identifies key challenges such as hydrogen embrittlement and permeability, especially at high concentrations and pressures. Blending hydrogen into natural gas networks emerges as a promising and cost-effective solution compared to building new hydrogen pipelines. If proven safe and efficient, this approach could reduce investment and delivery costs while accelerating hydrogen adoption. As described in [14], it serves as a transitional pathway that leverages current infrastructure. Their study highlights critical technical factors including material compatibility, pipeline integrity, and regulatory requirements, all of which must be addressed for successful implementation.

Complementing earlier work, [15], through contributions by San Marchi and Somerday, offers a detailed technical reference on materials compatibility with gaseous hydrogen. The report compiles academic and industry sources, focusing on hydrogen permeation and its effects on material integrity, grouped under "hydrogen-assisted fracture." It is organized by material types, including Plain Carbon Ferritic Steels, Low-Alloy Ferritic Steels, and Austenitic Steels. The section on Plain Carbon Ferritic Steels is especially relevant for pipelines due to their widespread use and mechanical behavior in hydrogen environments. The reference examines hydrogen's impact on tensile properties, fracture toughness, and fatigue behavior. These insights are crucial for evaluating the conversion of natural gas pipelines and guiding material selection for future projects. It also provides data on material composition and hydrogen exposure effects, helping operators assess how microalloying elements respond to hydrogen blends or pure hydrogen. Currently, hydrogen pipelines in the United States span about 2,600 km, mostly built with API 5L Grade X52 or lower, with a maximum diameter of 20 inches and a design factor of 0.5, compared to 0.72 for subsea pipelines [16]. Desktop studies are underway to expand material grades, pipe sizes, and improve design factors [16].

The case study [17] investigated the options and challenges associated with repurposing natural gas networks for hydrogen transport through a case study in Germany. Their study examined three approaches: enhanced maintenance without pipeline modification, the use of inhibitors, and a pipe-in-pipe configuration. Notably, all evaluations were conducted using pure hydrogen rather than a hydrogen-natural gas blend.

Subsea Pipeline Considerations

This paper also addresses specific requirements for subsea pipelines, which need to be designed to sustain unique operational and environmental challenges. These include:

- Hydrogen Diffusion into Water
- Hydrogen-Assisted Cracking (HAC):
- Integrity Monitoring Using In-Line-Inspection Tool (ILI)
- Conservative Design

These factors help assess subsea infrastructure suitability and needed design updates. The inherent conservatism within existing standards and guidelines for subsea hydrogen pipeline design will be highlighted to illustrate the conservative nature of current design practices, without engaging in critical analysis. For deeper insight, relevant literature is referenced.

Case Studies

To support the methodology and objectives of this paper, relevant case studies and pilot projects will be included to show how elements such as standards evaluation, material compatibility, and subsea considerations are applied in practice. Real-world examples help contextualize the repurposing of natural gas pipelines for hydrogen service and reveal gaps that must be addressed before implementation. In the offshore oil and gas sector, this concept is still emerging, making it essential to track ongoing projects and studies to stay informed about challenges, best practices, expected outcomes, and lessons learnt.

Future Envisaged Work

The methodology includes a forward-looking element by identifying research gaps and proposing future directions. Key areas include long-term field trials to validate hydrogen pipeline performance, development of advanced in-line inspection tools for hydrogen service, and selection of materials specifically suited for hydrogen transport. Refining material composition may also be necessary to improve compatibility, ensuring safety, reliability, and performance in hydrogen environments.

Results and Observations

Hydrogen Transport

According to [18], no single hydrogen transport method dominates across all applications. Instead, the optimal choice depends on national energy profiles, industrial needs, and infrastructure readiness. Pipelines are the most cost-effective for transporting large volumes of compressed hydrogen (CGH₂) over long distances, particularly when existing natural gas pipelines can be repurposed. Liquid hydrogen (LH₂) is ideal for shipping due to its high energy density, although liquefaction and boil-off losses remain technical challenges. LOHCs and ammonia (NH₃) offer logistical advantages, but both require significant energy for synthesis and cracking.

A similar conclusion was reached in [19] through a techno-economic modeling approach. Their analysis found that CGH₂ is cost-effective for short distances but gets increasingly expensive over longer routes due to low transport capacity and high energy losses. LH₂ trucks and CGH₂ pipelines were identified as the most economical for long-distance inland transport, while LH₂ shipping emerged as the cheapest option for overseas delivery. Ammonia was the most expensive carrier due to high costs of synthesis and cracking, with total energy losses reaching up to 35.8%. LOHCs, while logistically flexible and free from boil-off losses, suffered from high conversion energy demands (~37.3%) and low hydrogen content, making them less favorable overall. Combined transport pathways such as LH₂ shipping followed by LH₂ truck or pipeline were found to be the most cost- and energy-efficient.

Comparable findings were presented in [20] suggesting that no universal hydrogen carrier or transport mode is optimal across all regions. Instead, the choice should be tailored to project size, transport distance, and infrastructure maturity. Pipelines are most economical for short distances and large volumes, especially when repurposed infrastructure is an option, extending viability up to 8,000 km. Ammonia is the most promising for long-distance shipping due to its high energy density, existing global trade infrastructure, and potential for direct use in industrial and maritime applications. However, reconversion losses (13–34%) and high-temperature cracking requirements remain barriers. LOHCs offer compatibility with existing oil infrastructure and stable transport conditions but face challenges in carrier cost, scalability, and energy efficiency. IRENA's modeling projects that by 2050, ammonia shipping will likely dominate intercontinental

hydrogen trade, while pipelines will remain optimal for regional transport. The report includes decision matrices and cost-performance comparisons to support strategic infrastructure planning.

A techno-economic analysis by the IEA suggests that for transport distances exceeding 300–500 km at scale, pipelines are the most cost-competitive option compared with trucking or liquefaction [21]. Beyond techno-economic considerations, infrastructure strategy plays a critical role in shaping hydrogen transport decisions. Pipelines ensure supply security through continuous flow, reducing reliance on intermittent shipments. Their buffering capacity helps balance fluctuations in hydrogen production from renewable sources, stabilizing supply and demand. Pipelines also integrate seamlessly with existing energy networks, enabling hybrid systems where hydrogen complements electricity grids and natural gas infrastructure. For countries with extensive gas networks, repurposing or adapting pipelines accelerates the energy transition while minimizing capital investment and environmental impact [22], [23].

Collectively, these studies underscore the importance of tailoring hydrogen transport strategies to specific regional and project conditions, rather than pursuing a universal solution.

Standards and Guidelines Evaluation

Existing Standards. A range of national and international organizations have developed hydrogen-specific pipeline standards. Key examples include ASME B31.12 (USA), the first dedicated hydrogen pipeline code; ISO/TC 197, which oversees hydrogen technology standards but lacks a transmission-specific code; CGA G-5.6 (Canada), focused on industrial hydrogen piping; IGEM/TD/1 (UK), adapted for hydrogen blends; DVGW G 409 (Germany), addressing hydrogen-natural gas mixtures; and CEN initiatives under the European Green Deal, which aim to establish harmonized standards across Europe.

Identified Gaps. Despite progress in standard development, key gaps persist. In terms of materials, most existing pipeline codes impose conservative limits on steel grades, typically restricting the use to API 5L X52 or lower, due to concerns over hydrogen embrittlement [15]. While higher-strength steels are common in modern gas transmission systems, they currently lack well-established qualifications for hydrogen service.

Another aspect often overlooked in most regulatory frameworks is the probability of hydrogen diffusion into pipeline steel [24]. These frameworks assume that molecular hydrogen (H_2) has already dissociated into atomic hydrogen (H) and diffused into the steel. While this assumption is conservative and intended to prioritize safety, it may not accurately reflect the conditions present in transmission pipelines, where the energy required for dissociation is rarely encountered. Notwithstanding, atomic hydrogen can also be generated at actively growing fatigue cracks, where newly exposed metal surfaces facilitate the dissociation process. This raises concerns about whether atomic hydrogen can interact with mid-wall or external defects under typical operating conditions. Current understanding suggests that hydrogen entry through steel's oxide films is kinetically limited at standard pipeline temperatures. Nevertheless, limited research has been conducted to determine whether dissociation and entry could occur over the pipeline's lifetime [24].

Furthermore, due to its role in accelerating fatigue crack growth and reducing the fracture toughness of pipeline steels, ASME B31.12 mandates the implementation of fracture control plans. However, the predictive models currently available are based on limited experimental data. Ongoing research projects aim to bridge this gap by refining crack arrest models and enhancing the reliability of fracture assessments in hydrogen service [25], [26].

While ASME B31.12 covers hydrogen blends with 10% or more hydrogen, some of its safety calculations like impact radius (how far damage can spread) and arrested crack length (how far a crack travels before stopping) are based on pure hydrogen (100%). This becomes especially important when most of the mixture is natural gas, because the behavior of the blend can be quite different [24].

Subsea pipelines receive limited attention compared to other environments, and there are very few guidelines specifically for hydrogen pipelines under the sea. DNV's H2Pipe initiative is one of the first

efforts to study key factors such as cathodic protection, hydrogen diffusion, and fracture resistance in subsea conditions [25], [26].

One of the most pressing issues is the absence of standardized methodologies for evaluating hydrogen compatibility in vintage pipelines, which often lack detailed records of material composition, fabrication history, and operational conditions [27]. Discrepancies in strain rate specifications across standards such as those between ANSI/CSA and ASTM G142 contributes further to the inconsistent assessments of hydrogen embrittlement susceptibility [27]. In these tests, the specimen is exposed to a hydrogen environment and strained slowly until failure. A slower strain rate allows more time for hydrogen atoms to diffuse into the metal, increasing the likelihood of embrittlement, whereas a faster strain rate may lead to false-negative results by masking the material's susceptibility. Evaluation criteria like Relative Notched Tensile Strength (RNTS) and Relative Reduction of Area (RRA) are not totally applicable and may not be reliable in predicting fatigue or fracture performance under hydrogen exposure [27]. This is because RNTS and RRA are based solely on tensile testing without due consideration to cyclic loading.

Subsea Pipeline Considerations

At the time of writing, no dedicated subsea hydrogen pipelines are operational. However, the H2Pipe Joint Industry Project (JIP), currently in its second phase, aims to develop the world's first guidelines for transporting gaseous hydrogen through both existing and new offshore pipelines. The first phase of the JIP began in 2021 and ended in 2023. The second phase is primarily experimental, focusing on assessing the effects of hydrogen on pipeline materials [28].

Subsea hydrogen pipelines present unique integrity challenges due to the marine environment and hydrogen's chemical behavior. Hydrogen can diffuse into seawater, potentially interfering with cathodic protection systems and increasing corrosion risks. Hydrogen-assisted cracking (HAC), driven by combined mechanisms of stress corrosion cracking and hydrogen embrittlement, is particularly concerning under cathodic overprotection. Monitoring pipeline integrity is also more complex, as conventional inline inspection (ILI) tools may not detect hydrogen-specific damage modes like hydrogen attack, which often lacks magnetic flux leakage signatures. Due to limited operational experience, current subsea hydrogen pipeline designs adopt conservative safety margins, including reduced operating pressures and restricted material selections.

Material Compatibility and Mechanical Behavior

Despite its significant promise as a clean energy carrier and a potential substitute for fossil fuels, hydrogen presents unique challenges, chiefly related to material compatibility. These issues become more pronounced when existing natural gas infrastructure is repurposed for hydrogen service. Although hydrogen is transported in its stable molecular form (H_2), it can dissociate into atomic hydrogen at the surface of metallic pipelines, particularly in areas with welds, stress concentrations, or catalytic conditions [29]. This atomic hydrogen readily diffuses into the steel, especially low-carbon grades, leading to a degradation mechanism known as hydrogen embrittlement (HE), which compromises the structural integrity of the pipeline [29]. Hydrogen embrittlement (HE) is influenced by several factors, including hydrogen concentration and operating pressure [29]. Furthermore, the susceptibility of pipeline steels to HE is strongly dependent on the grade of steel and its mechanical properties, such as strength and hardness. This vulnerability applies not only to the base metal but also to welds and heat-affected zones (HAZ), where microstructural changes and residual stresses can promote hydrogen diffusion and trapping, increasing the likelihood of crack initiation and brittle failure [29]. This leads to accelerated fatigue crack growth under cyclic loading and slow crack growth near welds and stress zones, with cracks initiating from much smaller flaws than in natural gas environments [29].

Hydrogen introduction has a negligible effect on the tensile strength of pipeline steels; however, its impact becomes significantly more pronounced in ductility, fracture toughness, and fatigue crack growth

rate (FCGR) [30]. These adverse effects are especially critical under room temperature conditions, where hydrogen embrittlement mechanisms are most active [30]. Key findings from [30] indicate that no lower limit has been identified below which hydrogen exposure can be considered harmless to material integrity. In fact, even a 1% hydrogen concentration in a methane blend at 10 MPa resulted in an approximately 30% reduction in fracture toughness and tenfold increase in FCGR [30]. Under pure hydrogen conditions, fracture toughness was reduced by approximately 50% [30].

Modern natural gas pipelines use crack arrest models that are based on test data from Charpy V-notch (CVN) and drop-weight tear tests (DWTT). However, these methods don't work well for hydrogen pipelines because hydrogen tends to cause more brittle fractures. To address this, new testing methods are being developed through initiatives like PRCI and DNV's H2Pipe [25], [26].

Case Studies and Bridging Initiatives

One major issue highlighted in [17] is the high electricity demand of hydrogen compressors, which is much greater than that of natural gas compressors. Among the three repurposing options studied, using gas additives was the most cost-effective. However, public opinion and safety concerns may influence the final choice. Due to uncertainty in the hydrogen market, some operators might prefer low-cost solutions with minimal changes such as keeping the pipeline as-is but increasing maintenance.

Multiple real-world projects show that repurposing natural gas pipelines for hydrogen is technically achievable. Many operators are doing so after some assessment of the pipeline (to be converted) condition. Below are some of the examples:

In the Netherlands, Gasunie successfully converted a 12-km pipeline to carry 100% hydrogen, supplying industrial users. The project offered valuable lessons in flow behavior, material performance, and monitoring needs [31].

In the UK, HyNet North West initiative is testing hydrogen blending and preparing for full hydrogen service using existing transmission pipelines. The said initiative will transport hydrogen from Stanlow to industrial users and blending points at Partington and Warburton, integrating with the existing gas network. It will also connect to hydrogen storage facilities to balance supply and demand. The project includes approximately 125 km of underground high-pressure steel pipeline, user connection spurs, and several Above Ground Installations along the route [32].

The API 1104 Hydrogen Task Group conducted a detailed desktop study to assess the feasibility of extending hydrogen pipeline design beyond traditional U.S. standards [16]. Historically, hydrogen pipelines have been limited to API 5L Grade X52 or lower, diameters of 20 inches or less, and design factors not exceeding 0.50. With growing demand for hydrogen infrastructure, this study explored the use of higher-grade steels and more aggressive design parameters [16]. The focus was on Scenario 1, which evaluated hydrogen pipelines using pipe grades X52, X60, X65, and X70 at a diameter of 20 inches and a maximum allowable operating pressure (MAOP) of 1,440 psi. Two design factors were considered: 0.60 and 0.72. The study assessed axial stress ranges from 30% to 90% of specified minimum yield strength (SMYS) and fracture toughness values (K_{IH}) of 65 and 100 MPa $\sqrt{m}$. Hydrogen embrittlement was identified as a key concern, particularly affecting internal surface and near-surface flaws. Traditional flaw acceptance criteria (3×25 mm for surface flaws and 3×50 mm for buried flaws) were compared against Engineering Critical Assessment (ECA) criteria (5×50 mm). The study confirmed that pipelines constructed with higher-grade steels, such as X70, can tolerate these flaws if qualified to a K_{IH} of 100 MPa $\sqrt{m}$ [16]. ECA procedures were applied using API 1104 for girth welds and BS 7910 for seam welds, with and without residual stress considerations. The results demonstrated that both girth and seam welds in X52–X70 grades are viable under the evaluated conditions, supporting the safe expansion of hydrogen pipeline design [16].

To address standardization gaps in hydrogen pipeline infrastructure, the industry is advancing through collaborative initiatives. Notable among these is the H2Pipe Joint Industry Project, led by DNV, which is developing design standards for subsea hydrogen pipelines with contributions from operators, suppliers, and

major oil companies. The PRCI Hydrogen Task Force is conducting research on hydrogen embrittlement, fracture mechanics, and the repurposing of natural gas infrastructure, and these efforts are expected to be reflected in future revisions of ASME codes. Meanwhile, the European Hydrogen Backbone (EHB), although primarily focused on deployment, has sparked dialogue around harmonized European codes, especially for retrofitted assets. In the UK, government-supported programs such as HyNet will be generating critical safety data for hydrogen distribution and blending, which will guide revisions to IGEM standards. Collectively, these initiatives underscore the importance of coordinated action among regulators, standards organizations, and industry stakeholders to establish robust, globally harmonized guidelines for hydrogen pipelines.

Future envisaged work

Future work should focus on testing higher grade steels such as API 5L X70 in high pressure hydrogen environments to better understand their resistance to cracking and fatigue. This is especially important for subsea pipelines, where repairs are more difficult and expensive. At the same time, there is a need to develop inspection tools that work effectively with hydrogen, including technologies like acoustic sensors and chemical detectors. Real time monitoring systems can also help identify problems early and improve safety. To support international hydrogen transport, global standards should be aligned through cooperation between countries and organizations. Long term data from major projects such as the European Hydrogen Backbone and HyNet North West will be valuable for improving design and safety. It is also helpful for current and future projects to choose materials that are suitable for hydrogen use, so that future conversion is easier. Research should explore the actual risk of atomic hydrogen entering materials, rather than assuming it will always occur. Finally, subsea hydrogen pipelines should receive special attention, as they face different challenges compared to pipelines on land.

Conclusion

Hydrogen presents a viable pathway for achieving global decarbonization goals, and repurposing existing natural gas pipelines offers a cost-effective and scalable solution for its transport. However, this transition is not without challenges. The review of current standards and guidelines reveals significant gaps in material qualification, fracture control, and inspection methodologies, particularly for subsea applications. Hydrogen embrittlement, fatigue behavior, and fracture toughness remain critical concerns, especially under high-pressure and cyclic loading conditions. While blending hydrogen into natural gas networks offers a transitional strategy, the long-term goal of pure hydrogen transport demands more robust and tailored standards. There is currently no universal benchmark for determining the appropriate percentage of hydrogen in natural gas blends, which necessitates a case-by-case assessment of each pipeline before repurposing. However, multiple studies suggest that blends containing up to 17% hydrogen are generally considered acceptable for use in existing natural gas infrastructure, depending on material compatibility and operating conditions [33]. Collaborative initiatives such as H2Pipe, PRCI-ASME, and HyNet North West are paving the way by generating essential data and developing new design frameworks. Future efforts must focus on validating material performance through field trials, enhancing inspection technologies, and aligning international standards to support cross-border hydrogen infrastructure. Selecting hydrogen-compatible materials from the outset will simplify future conversions and improve safety and reliability. Ultimately, a coordinated approach involving regulators, industry stakeholders, and research institutions is essential to establish a globally harmonized framework for hydrogen pipeline transport.

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